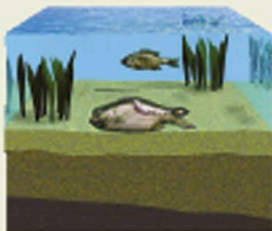


Studying fossils

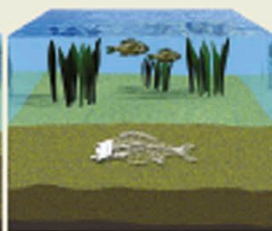
Fossils are the remains of plants or animals that once lived on Earth and have been preserved in rocks for millions of years. It was in the 1600s that scientists first began to wonder how fossils had been formed. By the 19th century, paleontology (the study of fossils) had become a recognized science. Today, paleontologists are able to study the DNA (see pp.198–199) of some fossils to give us a clearer picture of life on Earth.

How fossils form

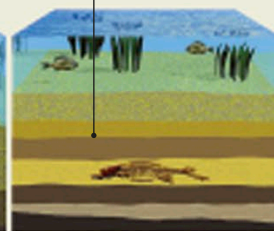
Usually when an organism dies, its remains rot and disappear. For the remains to become a fossil, the conditions have to be just right. It helps if the plant or animal is living in a watery environment and is quickly buried in mud or sand after death.



1. The body of a dead fish falls to the bottom of the ocean. It quickly sinks into the mud or sand.



2. The soft parts of the body rot, leaving the hard bones behind. More sediment falls through the ocean and covers the bones.



3. The fish's skeleton dissolves and is replaced by minerals that harden, while the sediment around it turns to rock.

More layers of sediment build up

Types of fossil

There are two types of fossil.

A body fossil is formed from the hard parts of a plant or animal's structure—woody trunks, shells, teeth, or bones. A trace fossil is something left behind by an animal, such as its footprints, eggs, or dung.



Insect in amber

Trapped in amber

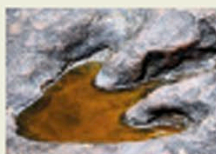
Millions of years ago this spider was trapped in resin, a sticky liquid that oozes out of some trees. The resin fossilized into amber, preserving the body of the spider intact.



Ammonite fossil

Turned to stone

Most body fossils are copies, or molds, of the original organism. As the bones, or shell in the case of this ammonite, dissolved, minerals seeped in to fill the space left behind in the mud or sand. Gradually, the minerals hardened into stone.



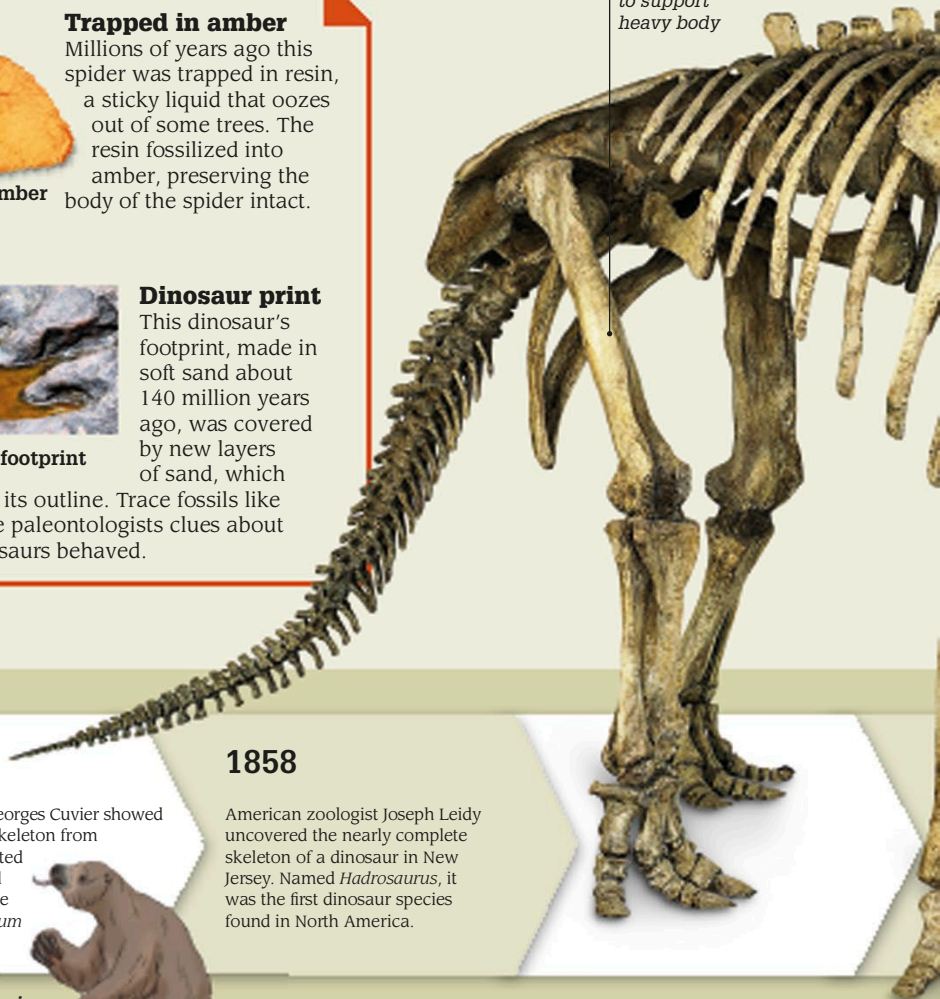
Dinosaur footprint

Dinosaur print

This dinosaur's footprint, made in soft sand about 140 million years ago, was covered by new layers of sand, which

protected its outline. Trace fossils like these give paleontologists clues about how dinosaurs behaved.

Strong legs to support heavy body



Key events

1669

Danish naturalist Nicolas Steno suggested that fossils are the remains of once-living creatures deposited in layers of sedimentary rock that formed slowly over time.

1796

French zoologist Georges Cuvier showed that a giant fossil skeleton from Argentina was related to the modern land sloth. He named the creature *Megatherium* (giant beast).

1858

American zoologist Joseph Leidy uncovered the nearly complete skeleton of a dinosaur in New Jersey. Named *Hadrosaurus*, it was the first dinosaur species found in North America.

Megatherium

